MIRG
Mortality and Inequalities Research Group

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Kinship demography I

How demographic rates shape kinship networks

Yu Chen & José Manuel Aburto

06/03/2026

Learning objective

By the end of this lecture, you will understand

1. how **fertility** and **mortality** determine the expected number of relatives across the life course;
   - kinship network
   - bereavements (parental & grandparent caregivers loss);

2. how to quantify and interpret **inequality** of orphanhood across race/ethnicity.

Recap

What have we learnt so far?

Population growth; Fertility estimation; Mortality estimation.

Now we move our research interests to mortality outcomes in the family …

Outline

Part I: Why kinship matters and the definition

Part II: Demographic foundation + R lab

Part III: Discussion of the paper on ``orphanhood and grandparent caregiver loss by cause of death'' estimation

Part IV: Orphanhood inequality analyses

Part I: Motivation

Why kinship matters?

Consider a woman born in Sweden in 2018.

At age 50:

- What is the probability that her mother is still alive?
- How many living grandparents did she likely have at birth?
- How many living siblings might she have on average?
- How many children might she have on average?

Why kinship matters

- What is the probability that her mother is still alive?
- How many living grandparents did she likely have at birth?
- How many living siblings might she have on average?
- How many children might she have on average?

Can you answer these questions with fertility or mortality alone?

Demography <=> relatives

- Demography is experienced through relatives

Can you think some examples to support this statement?

How many grandparents, siblings, cousins are in your family?

Def: Kinship and kinship demography[1]

Kinship: Social relationships that bind individuals together through culturally shared definitions of relatedness on biological, legal, or normative grounds, ultimately constituting family systems.

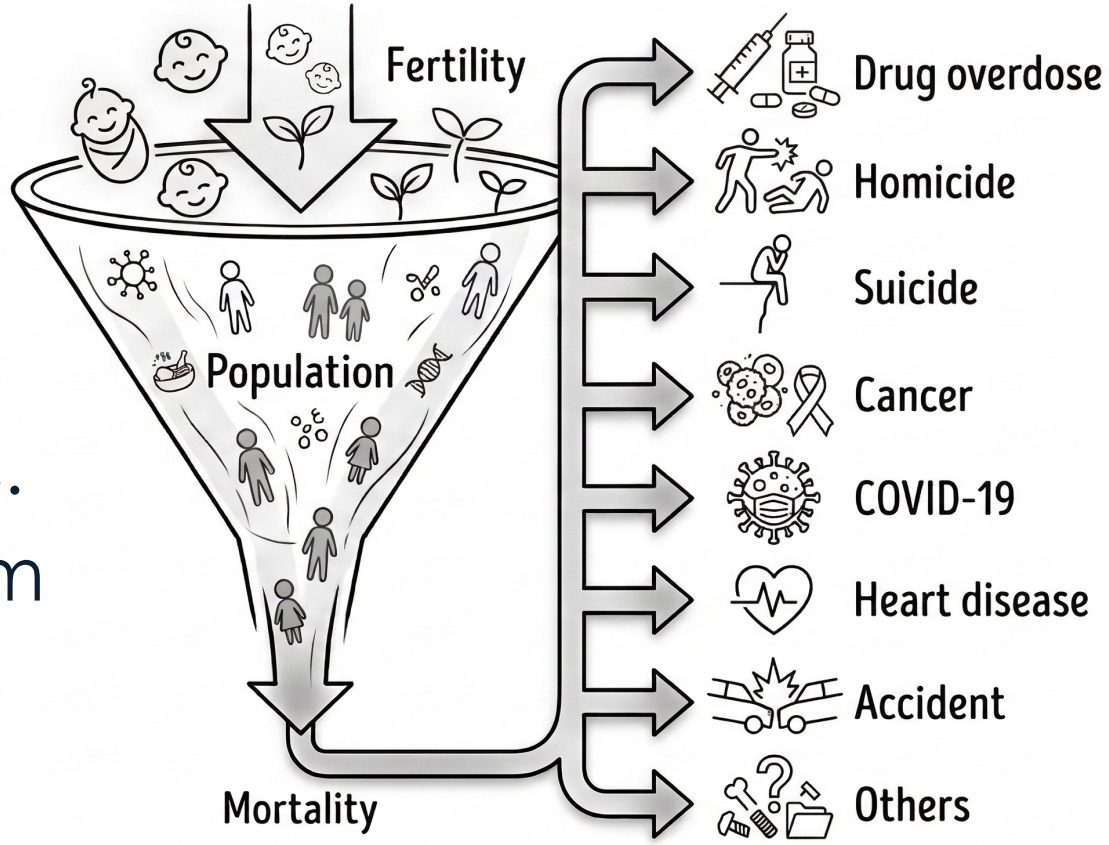
Kinship demography studies *kinship networks* and their *dynamics* using demographic methods.

[1] Alburez-Gutierrez, Diego, et al. "Kinship, demography, and inequality: Review and key areas for future development." *Preprint. SocArXiv. https://doi.org/10.31235/osf. io/fk7x9* (2022).

Kinship demographic rate

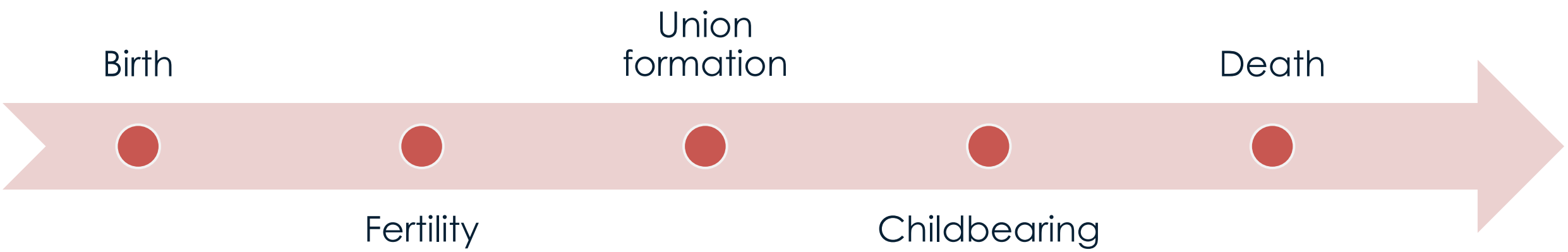
Fertility and mortality rates affect kinship network:

- Fertility produces new individuals.
- Mortality removes individuals from the population.



Generated by Google Gemini Nano based on the author's instructions.

Part II: Demographic Foundations



Kinship networks evolve across the life course.

How timing will affect the kinship network?

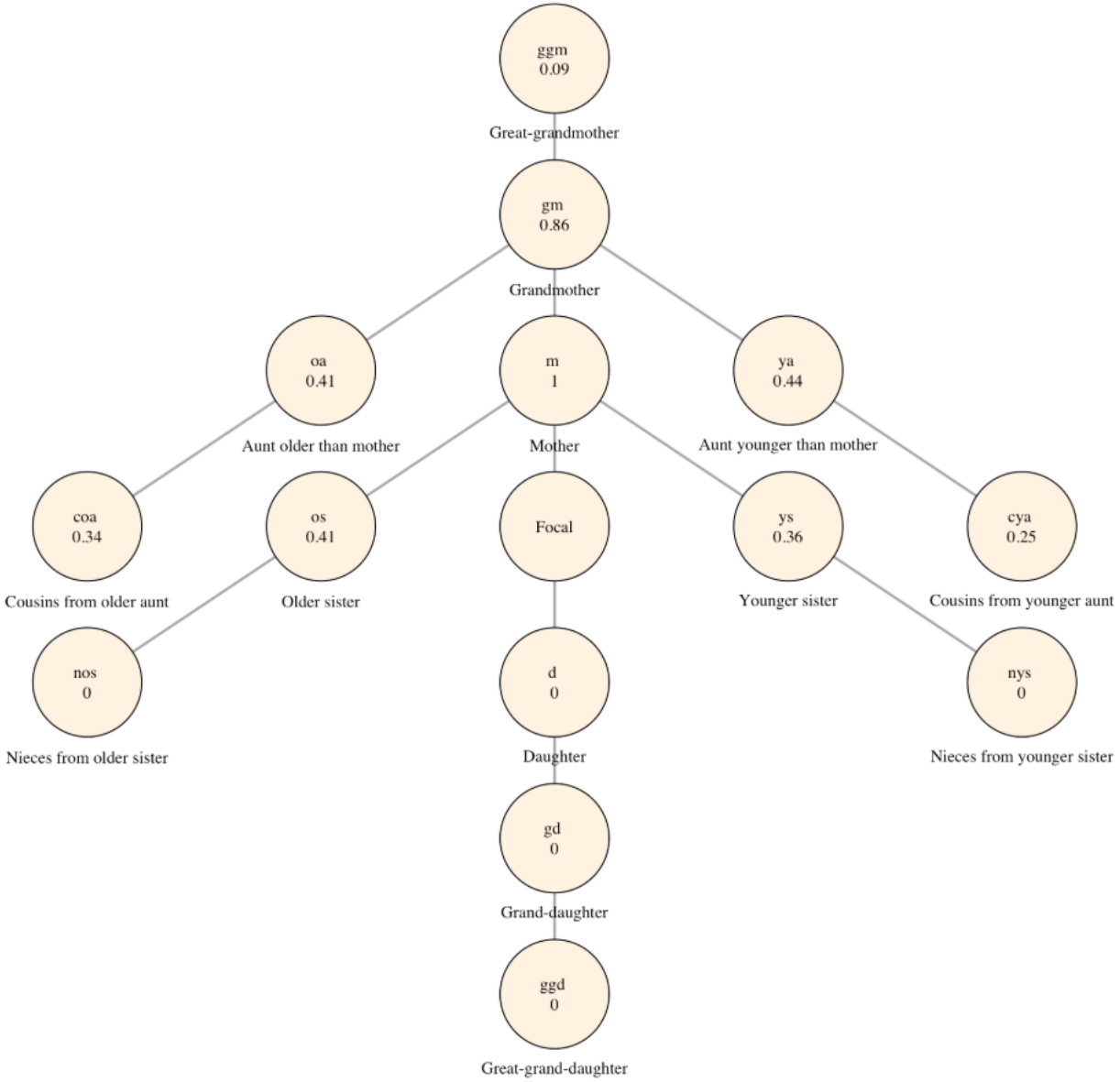
Kinship network

How can we estimate how many relatives *a focal* will have?

There is a useful package DemoKin

https://ivanwilli.github.io/DemoKin/

Let's coding!



Formal kinship matrix, please check: Caswell, Hal. "The formal demography of kinship." *Demographic Research* 41 (2019): 679-712.

Hands on exercise

DemoKin

Installation
Load libraries
Kinship Network
Age profiles of living kin

This is an R package for the demographic analysis of kinship networks using matrix-based models.
It implements methods developed initially by Caswell (2019), and continued in successive papers (like Caswell (2020) or Caswell (2024)) for estimating the number and age distribution of relatives under various demographic assumptions.

Features

- Estimate kin counts and age distributions for various types of relatives
- Support for one-sex and two-sex models
- Time-invariant and time-varying approaches
- Multi-state models incorporating additional variables like parity or education
- Visualization tools for kinship networks

Installation

You can install the development version of DemoKin from GitHub:

```
# install.packages("remotes")
remotes::install_github("IvanWilli/DemoKin")
library(DemoKin)
```

https://ivanwilli.github.io/DemoKin/

Get the html for R code in GitHub (click)

Kinship demography practical demo

Yu Chen

2026-03-05

Installation

```
# install.packages("devtools")
devtools::install_github("IvanWilli/DemoKin")
```

Load libraries

install.packages("DemoKin")
require(DemoKin)

```
require(DemoKin)
require(dplyr)
require(tidyr)
require(ggplot2)
```

Kinship Network

The R practical materials are adapted from https://alburez.me/EDSD_2025_kinship/

Foundation paper

ELSEVIER

Theoretical Population Biology

Volume 5, Issue 1, February 1974, Pages 1-27

Family formation and the frequency of various kinship relationships

Leo A. Goodman, Nathan Keyfitz, Thomas W. Pullum

Show more

+ Add to Mendeley Share Cite

https://doi.org/10.1016/0040-5809(74)90049-5

Get rights and content

Abstract

A set of age-specific rates of birth and death implies expected numbers of kin. An individual girl or woman chosen at random out of a population whose birth and death rates are specified can be expected to have a certain number of older sisters, younger sisters, nieces, cousins; expressions for these values are provided for both total kin and kin who are still living. Included also are the probabilities of living mother, grandmother, and great grandmother for girls and women of various ages. The methods are applicable to the size of the nuclear and the extended family. All formulas have been programmed and specimen numerical values are given.

Demographic Foundations of kin structure

Advanced materials

- Pipeline to compute the kinship matrix and bereavement

If you are interested in advanced method to compute the formal kinship matrix (including the matrix calculation…)

Caswell, Hal. "The formal demography of kinship." Demographic Research 41 (2019): 679-712.
Caswell, Hal. "The formal demography of kinship IV." Demographic Research 47 (2022): 359-396.

Advanced materials

1. Leslie matrix — population projection

$$\begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}(t+1) = \left[\underbrace{\begin{pmatrix} 0 & 0 & 0 \\ p_1 & 0 & 0 \\ 0 & p_2 & 0 \end{pmatrix}}_{\mathbf{U}} + \underbrace{\begin{pmatrix} f_1 & f_2 & f_3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{\mathbf{F}} \right] \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}(t)$$

$$\mathbf{n}(t+1) = (\mathbf{U}+\mathbf{F})\,\mathbf{n}(t)$$

$\mathbf{U}$ = survival matrix

$\mathbf{F}$ = fertility matrix

Advanced materials

2. Formal kinship matrix — surviving kins and the subsidy

For each type **k** of kin:

$$\begin{aligned} \mathbf{k}(x+1) &= \mathbf{U}\mathbf{k}(x) + \beta(x) \\ \mathbf{k}(0) &= \mathbf{k}_0 \end{aligned}$$

$$\begin{aligned} \mathbf{k}(x) &= \text{age distribution of some type of kin at age } x \text{ of Focal} \\ \mathbf{U} &= \text{survival matrix} \\ \\ \beta(x) &= \text{recruitment 'subsidy' at age } x \text{ of Focal} \\ &= \begin{cases} \mathbf{0} & \text{no recruitment} \\ \mathbf{F}\,\mathbf{k}^*(x) & \text{recruitment from kin of type } \mathbf{k}^* \end{cases} \\ \\ \mathbf{k}_0 &= \text{initial condition (kin at the birth of Focal)} \end{aligned}$$

Advanced materials

3. Extension: death and bereavement

$$\widetilde{\mathbf{k}}(x) = \left(\frac{\mathbf{k}_{\text{living}}}{\mathbf{k}_{\text{dead}}} \right)(x)$$

Cumulative deaths up to age x

$$\widetilde{\mathbf{k}}(x+1) = \left(\begin{array}{c|c} \mathbf{U} & \mathbf{0} \\ \hline \mathbf{M} & \mathbf{I} \end{array} \right) \widetilde{\mathbf{k}}(x) + \widetilde{\beta}(x)$$

where

$$\mathbf{M} = \text{diag}(\mathbf{q})$$

Break

Q & A

Yu Chen

Research Fellow

London School of Hygiene and Tropical Medicine

Yu.Chen@lshtm.ac.uk

https://www.lshtm.ac.uk/aboutus/people/chen.yu

Lecture notes and R labs:
https://github.com/YuCHENJT/PDP_2025-2026_lecture_materials